

Learning High-Precision Contact Skills from Behaviour Priors: Variability-Driven Variable Impedance Control with State-Driven Task Parametrization, Stability and Passivity Guarantees

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Abstract

This report presents a framework for transferring a high-precision contact skill — cutting with a scalpel-like tool — to a robot manipulator by learning from human demonstrations (termed *behaviour priors*). Rather than replaying a single trajectory, the method learns the underlying structure of the skill and reproduces it under a variable-impedance controller with an implemented energy-tank passivity mechanism at the joint torque port.

The contribution comprises three elements. First, a *variable impedance controller* whose stiffness is not hand-tuned but learned from the consistency of the demonstrations: a quadratic program (QP) solved at each control step maps the demonstration precision onto the controller gains, so the manipulator is stiff only along directions that the demonstrations regulated and compliant elsewhere. Second, a *state-driven task parametrisation*: the motion is indexed by a progress (phase) variable whose evolution is corrected by the manipulator’s measured state rather than by a fixed clock, providing temporal-scale adaptability and disturbance rejection. Third, a formal *stability and passivity* analysis: the torque command is projected onto a live energy-tank power budget, and a positive-power perturbation validation demonstrates that the tank prevents energy injection that a controller without tank projection permits. Exponential convergence is derived under the stated regulation assumptions and is reported empirically as a diagnostic on the regulated task error rather than overclaimed from the released directions.

A key empirical observation motivates the design: across repeated demonstrations the tool *position* is highly repeatable (cross-trial correlation ≈ 0.95), whereas the *force* is not (≈ 0.10). The controller therefore regulates position and treats force as a constraint, while a separate per-material *cutting-force model*, identified from each material’s own data, provides the interaction force in simulation. The framework is evaluated on three materials (cork, PVC, penoplex) for both a fixed and a moving workpiece.

Contents

Nomenclature	2
1 Problem statement	3
2 What in the demonstration is actually a reliable skill?	3
3 Variability-driven variable impedance control	4
3.1 The impedance control law	4
3.2 Gains learned from demonstration precision	4
3.3 The online quadratic program for the gains	5

4	State-driven task parametrization	6
5	Stability and passivity	6
5.1	Passivity — the controller cannot create energy	7
5.2	Implemented torque-port passivity	7
5.3	Exponential stability — the error decays at a guaranteed rate	8
6	Material interaction model	8
7	Results	10
8	Novelty and contributions	10
9	Assumptions, scope and limitations	11

Nomenclature

The table below lists every symbol used, its meaning, and its physical dimension. Bold lower-case symbols are 3-vectors expressed in the world frame; n is the number of robot joints ($n = 7$ for the KUKA iiwa 14 used here).

Symbol	Meaning	Units
ϕ	task phase (progress along the skill), 0 at start, 1 at end	dimensionless
x, x_d	actual / desired tool-tip (TCP) position	m
$\dot{x}, \dot{x}_d$	actual / desired tool-tip velocity	m s^{-1}
$\tilde{x} = x_d - x$	position tracking error	m
$\dot{\tilde{x}} = \dot{x}_d - \dot{x}$	velocity tracking error	m s^{-1}
f	commanded Cartesian force (impedance “virtual force”)	N
f_{ext}	external (environment/contact) force on the tool	N
$K(\phi)$	diagonal stiffness matrix (per axis)	N m^{-1}
$D(\phi)$	diagonal damping matrix	N s m^{-1}
K_I	integral gain	$\text{N m}^{-1} \text{s}^{-1}$
ζ	damping ratio (set to 0.7)	dimensionless
Λ, M	task-space (Cartesian) inertia	kg
$C\dot{q}, g$	Coriolis/centrifugal and gravity joint torques	N m
τ	commanded joint torques	N m
$q, \dot{q}$	joint angles / velocities	rad, rad s^{-1}
J	TCP Jacobian ($\dot{x} = J\dot{q}$), $3 \times n$	m rad^{-1}
λ^2	damped-least-squares regulariser	$\text{m}^2 \text{rad}^{-2}$
N	null-space projector ($I - J^+ J$)	dimensionless
$\sigma_i^2(\phi)$	cross-trial variance of (range-normalised) signal, axis i	dimensionless
$w_i(\phi)$	precision weight (normalised inverse variance) $\in [0, 1]$	dimensionless
$K_i^{\text{pos}}(\phi)$	precision-derived position stiffness, axis i	N m^{-1}
$K_{\text{floor}}, K_{\text{min}}, K_{\text{max}}$	stiffness floor / box limits	N m^{-1}
$q_f(\phi)$	force-consistency weight in the QP $\in [0, 1]$	dimensionless
F_d	desired interaction force (from the prior)	N
F_i^{bound}	per-axis demonstrated force limit	N
$\hat{\phi}$	state-estimated phase	dimensionless
ϕ_{nom}	nominal (clock) phase rate	s^{-1}

Symbol	Meaning	Units
k_ϕ	phase-correction gain $\in [0, 1]$	dimensionless
ρ_x, ρ_F	normalising scales for position / force matching	m, N
V	Lyapunov / storage function (an energy)	J
T	energy-tank state	J
P_{inj}	power drawn by time-varying gains / moving reference	W
β	Lyapunov cross-term coefficient	s^{-1}
α, λ	exponential decay rates	s^{-1}
d	blade penetration depth below the material surface	m
k_{mat}	material loading stiffness	N m^{-1}
$F_{\text{mat}}^{\text{cut}}$	material cutting-force plateau	N

1 Problem statement

We want a robot to acquire a contact-rich manipulation skill, cutting, from a handful of human demonstrations. Each demonstration provides, sampled over time, the tool pose, its twist (linear/angular velocity), and the interaction wrench measured at the tool. Our aim is not to replay one recording but to capture the *skill*: a controller that performs the task accurately, respects the force character of the demonstrations, stays stable and safe in contact, and generalises across execution speeds and across materials.

The core difficulty is that, during contact, position and force are not independent. In a direction where the tool presses on the material, fixing the position determines the force and vice-versa; one cannot command both freely. A principled controller must therefore *decide what to regulate in each direction*, and a credible simulation must *model the material* that gives rise to the contact force in the first place. Both points are addressed below.

2 What in the demonstration is actually a reliable skill?

We first determine which components of the demonstration are sufficiently repeatable across trials to be treated as the skill. For each signal we measure, across trials, (a) the *shape correlation* (how similarly the signal evolves from trial to trial after removing its mean) and (b) a *regulation score* — the fraction of the signal’s variance that is explained by the task phase, i.e. a phase-locked signal-to-noise ratio.

Quantity	cross-trial shape correlation	regulation score
Position (X/Y/Z)	0.92–0.98	up to 0.73 (Z)
Force (F _x /F _y /F _z)	0.04–0.27	≈ 0

Table 2: How repeatable each quantity is across demonstrations (cork $n=28$, PVC $n=33$, penoplex $n=22$, 300 samples per trial).

The result is clear: *the demonstrated tool path is reproducible; the demonstrated force is not*. The recorded force is dominated by a large, nearly constant offset (consistent with tool preload / sensor bias) plus trial-to-trial content that does not repeat; this was checked against the dynamic part of the signal, time alignment and sign convention. The practical implication is that *point-tracking the demonstrated force trajectory is not a well-posed objective*. This is exactly the situation the *minimal-intervention principle* addresses: regulate strongly where the demonstration is consistent (position), and remain compliant where it is not (force) — the latter being treated as a constraint and as the interaction outcome of a calibrated material model.

3 Variability-driven variable impedance control

3.1 The impedance control law

The robot is controlled as a virtual mechanical impedance at the tool tip. The commanded Cartesian force is

$$f = K(\phi) \tilde{x} + K_I \int_0^t \tilde{x} d\sigma + D(\phi) \dot{\tilde{x}} \quad (1)$$

The terms are interpreted as follows:

- $K(\phi) \tilde{x}$ is a position-error spring. $K(\phi) [\text{N m}^{-1}]$ is a diagonal stiffness (one value per Cartesian axis) and $\tilde{x} [\text{m}]$ is the position error, so the product has units of force $[\text{N}]$. A larger K pulls the tool back to the reference more firmly.
- $K_I \int \tilde{x} d\sigma$ is an integral term that removes slow, steady-state position error; $K_I [\text{N m}^{-1} \text{s}^{-1}]$ multiplied by the time-integral of the error $[\text{m} \cdot \text{s}]$ again yields $[\text{N}]$.
- $D(\phi) \dot{\tilde{x}}$ is a velocity-error damper that suppresses oscillation; $D [\text{N s m}^{-1}]$ times $\dot{\tilde{x}} [\text{m s}^{-1}]$ gives $[\text{N}]$.

The damping is tied to the stiffness by a critical-damping rule, $D(\phi) = 2\zeta\sqrt{M K(\phi)}$, with damping ratio $\zeta = 0.7$ (a standard, slightly under-damped, fast and non-oscillatory response) and $M [\text{kg}]$ an effective mass. The virtual force is realised by joint torques through a damped-least-squares Jacobian transpose, with a posture task in the null space and model-based gravity/Coriolis compensation:

$$\tau = J^\top (J J^\top + \lambda^2 I)^{-1} f + N \tau_{\text{posture}} + g + C\dot{q}, \quad N = I - J^\top (J J^\top + \lambda^2 I)^{-1} J. \quad (2)$$

Here $J [\text{m rad}^{-1}]$ is the tool Jacobian ($\dot{x} = J\dot{q}$); $(J J^\top + \lambda^2 I)^{-1}$ is the damped pseudo-inverse that maps the Cartesian force to joint torques while remaining numerically robust near kinematic singularities (λ^2 is a small regulariser); N projects the secondary posture torque τ_{posture} into the null space so it does not disturb the task; and $g + C\dot{q} [\text{N m}]$ cancel gravity and Coriolis effects so that (1) governs the closed loop.

3.2 Gains learned from demonstration precision

The novelty is that $K(\phi)$ is not chosen by hand — it is the *precision* (inverse variance) of the demonstrations. Intuitively, where the human was very repeatable, that direction matters and should be tracked stiffly; where they were not, the controller should be soft. For each phase ϕ and axis i we form a precision weight on one shared scale and map it to stiffness:

$$w_i(\phi) = \text{clip}\left(\frac{1/(\sigma_i^2(\phi) + \epsilon)}{p_{\text{ref}}}, 0, 1\right), \quad K_i^{\text{pos}}(\phi) = K_{\text{floor}} + (K_{\text{max}} - K_{\text{floor}}) w_i(\phi). \quad (3)$$

$\sigma_i^2(\phi)$ is the cross-trial variance of the range-normalised signal at phase ϕ (dimensionless, so axes are comparable); ϵ is a tiny constant avoiding division by zero; p_{ref} is a robust 95th-percentile normaliser *shared across all axes and channels*, so a genuinely imprecise direction receives a genuinely small weight $w_i \rightarrow 0$. The weight $w_i \in [0, 1]$ then scales the stiffness linearly between a floor $K_{\text{floor}} [\text{N m}^{-1}]$ (kept non-zero so the arm can always execute the motion) and the maximum K_{max} . Figure 1 shows the result: with no manual tuning, the cut-depth axis becomes stiff while the lateral and force directions stay compliant — precisely the structure predicted by Section 2.

The same procedure applied to the other two materials (Fig. 2) yields the *same qualitative structure* — depth regulated, force released — but with *material-specific magnitudes*, because the gains are computed from each material’s own demonstration. This is the gain-side counterpart of the per-material force law of Section 6: one principle, three distinct parameter sets.

Learned-variability gains — cork (n=28 demos)

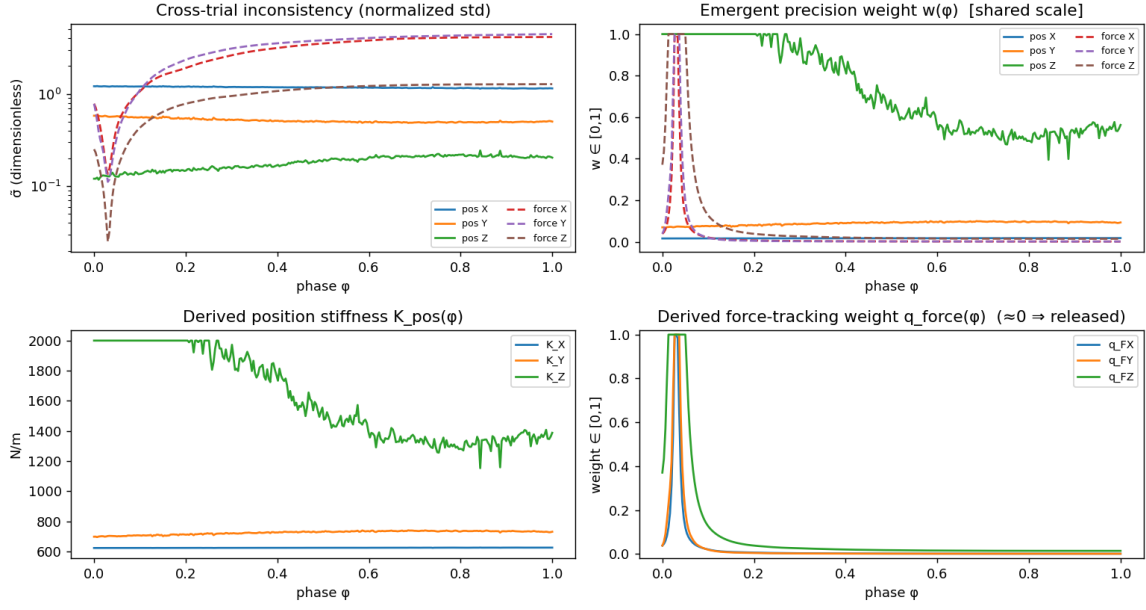


Figure 1: Gains learned from the cork demonstrations. Top-left: cross-trial inconsistency of each channel; top-right: the emergent precision weight $w(\phi)$; bottom: the resulting position stiffness $K^{\text{pos}}(\phi)$ and the (near-zero) force-tracking weight. The depth axis (blue) is regulated stiffly; force is released. *This split emerges from the data, not from tuning.*

3.3 The online quadratic program for the gains

At every control step the gains are obtained from a small convex quadratic program (QP) that *blends* two objectives and respects two constraints:

$$\begin{aligned} K^* = \arg \min_{K \in \mathbb{R}^3} & q_f(\phi) \|K \odot \tilde{x} - F_d\|_2^2 + (1 - q_f(\phi)) \|K - K^{\text{pos}}(\phi)\|_2^2 \\ \text{s.t. } & K_{\min} \leq K \leq K_{\max}, \quad -F_i^{\text{bound}} \leq K_i \tilde{x}_i \leq F_i^{\text{bound}} \quad \forall i. \end{aligned} \quad (4)$$

Term by term:

- $\|K \odot \tilde{x} - F_d\|_2^2$ is the *force-consistency* cost ($\odot$ is element-wise product): it asks the impedance force $K \odot \tilde{x}$ [N] to match the desired force F_d [N]. Its weight $q_f(\phi) \in [0, 1]$ is the *force-precision* weight — small, because force is not reproducible (Sec. 2).
- $\|K - K^{\text{pos}}(\phi)\|_2^2$ pulls the stiffness towards the precision-derived value K^{pos} [Nm⁻¹] of (3), weighted by $1 - q_f$. Because $q_f \rightarrow 0$, this term dominates and the QP returns $K \approx K^{\text{pos}}$: the controller tracks the precision-weighted position skill.
- The box constraint $K_{\min} \leq K \leq K_{\max}$ keeps the stiffness in a safe physical range.
- The *force-saturation* constraint $|K_i \tilde{x}_i| \leq F_i^{\text{bound}}$ guarantees the commanded force on each axis never exceeds the demonstrated force limit F_i^{bound} [N] — i.e. the desired forces act as *constraints*, as required.

The damping follows from K^* by the critical-damping rule. The QP is tiny (three variables) and solves in well under a millisecond (OSQP), so it runs in the real-time loop.

Controlled comparison: isolating the contribution of the learned variance structure.

To establish that the performance gain originates from the *specific* learned precision structure, rather than from gain adaptation per se, we evaluate the identical task and controller under four gain assignments: (i) the identified precision; (ii) *inverted* precision (regulating directions that were inconsistent and relaxing those that were consistent); (iii) precision *permuted* across axes; and (iv) a baseline that tracks the demonstrated force directly. The metric is the *cut-depth error* — the task-critical, demonstration-regulated axis — evaluated over 12 seeds with randomised contact parameters (Fig. 3, Table 3).

Gain assignment	cut-depth error (mm)	relative to identified	paired Wilcoxon p
Identified precision	10.9 ± 2.5	—	—
Permuted precision	14.7 ± 2.9	+26%	0.001
Inverted precision	20.7 ± 4.0	+47%	0.001
Direct force-tracking	20.9 ± 5.9	+48%	0.001

Table 3: The identified precision attains the lowest cut-depth error on all 11 stable seeds. Inverting or permuting the structure increases the error by 26–48% ($p = 0.001$, the smallest value attainable by the paired Wilcoxon test at this sample size). This isolates the identified *variance structure*, rather than gain adaptation in general, as the determinant of task performance.

4 State-driven task parametrization

Instead of being a function of time, the reference is a function of a *phase* variable $\phi \in [0, 1]$. This separates *what* the skill is from *how fast* it runs, which is what lets the same prior be executed at different speeds. The novelty here is *how* ϕ advances: not by a fixed clock, but by a correction driven by the robot’s *measured state*. At each step we estimate the phase $\hat{\phi}$ that best matches the current state to the learned trajectory over a short forward window, then correct ϕ towards it:

$$\hat{\phi} = \arg \min_{\varphi \in [\phi, \phi + \Delta]} \left[\frac{\|x_{\text{ref}}(\varphi) - x\|}{\rho_x} + \mathbb{K}_{\text{contact}} \frac{|\|F_{\text{ref}}(\varphi)\| - \|F\||}{\rho_F} \right], \quad \phi \leftarrow \text{clip}(\phi + \dot{\phi}_{\text{nom}} + k_{\phi}(\hat{\phi} - \phi)). \quad (5)$$

The bracketed cost compares where the robot *is* to where each candidate phase φ says it *should be*: the first term is the position mismatch $\|x_{\text{ref}}(\varphi) - x\|$ [m] normalised by a scale ρ_x [m]; the second, active only in contact ($\mathbb{K}_{\text{contact}}$), compares force magnitudes normalised by ρ_F [N] — force is the more informative cue once the tool is engaged. The update mixes the nominal clock rate $\dot{\phi}_{\text{nom}}$ [s^{−1}] with a proportional correction of gain $k_{\phi} \in [0, 1]$ towards the state estimate $\hat{\phi}$. Setting $k_{\phi} = 0$ recovers a pure time clock; with $k_{\phi} > 0$ the phase *tracks the robot’s actual progress* — it slows or pauses if the robot is held back and speeds up if it runs ahead. Searching only forward ($\varphi \geq \phi$) guarantees monotone progress, so the skill never regresses to an earlier stage. This is what gives the execution both speed-adaptability and robustness to disturbance.

5 Stability and passivity

We now establish the stability properties of the closed loop. The tool obeys the task-space dynamics $\Lambda \ddot{x} + C \dot{x} + g = f + f_{\text{ext}}$, where Λ [kg] is the Cartesian inertia and f_{ext} [N] the environment force. Substituting the control law (1) (with gravity/Coriolis cancelled by (2)) gives the *error dynamics*

$$\Lambda \ddot{\tilde{x}} + (C + D(\phi)) \dot{\tilde{x}} + K(\phi) \tilde{x} = f_{\text{ext}}, \quad (6)$$

a (time-varying) mass–spring–damper in the tracking error driven by the contact force.

5.1 Passivity — the controller cannot create energy

Proposition 1 (Continuous-time tank passivity). *At the Cartesian contact port, the input–output map $f_{ext} \mapsto \dot{x}$ is passive under the following tank dynamics. With the storage function (an energy, in joules) $V = \frac{1}{2}\dot{\tilde{x}}^\top \Lambda \dot{\tilde{x}} + \frac{1}{2}\tilde{x}^\top K \tilde{x} + T$ and a non-negative energy tank $T \geq T_{\min}$ whose dynamics harvest the power dissipated by damping and pay out the power injected by the time-varying gains and moving reference,*

$$\dot{T} = \underbrace{\dot{\tilde{x}}^\top D \dot{\tilde{x}}}_{\text{harvested [W]}} - \underbrace{P_{inj}}_{\text{[W]}}, \quad P_{inj} = \frac{1}{2}\tilde{x}^\top \dot{K} \tilde{x} + \tilde{x}^\top K \dot{x}_d, \quad (7)$$

the loop satisfies $\dot{V} \leq f_{ext}^\top \dot{x}$ as long as $T > T_{\min}$.

Interpretation: V is the total stored energy (kinetic + elastic + tank). The two indefinite, potentially energy-injecting terms — the rate of change of stored elastic energy from a rising stiffness ($\frac{1}{2}\tilde{x}^\top \dot{K} \tilde{x}$) and the power supplied to a moving reference ($\tilde{x}^\top K \dot{x}_d$), both in watts — are drawn from the tank, which is replenished by the power dissipated through damping. When the tank reaches its lower bound, stiffness growth is inhibited ($\dot{K} \leq 0$) and the phase is frozen ($\dot{\phi} \rightarrow 0$), so the controller cannot generate energy. This is the continuous-time passivity argument used to design the implementation.

5.2 Implemented torque-port passivity

The simulation controller enforces the tank constraint at the joint torque port, where the mechanical power supplied by the actuators is $P_\tau = \dot{q}^\top \tau$. At every control step the nominal torque

$$\tau_{\text{nom}} = J^\top f + \tau_{\text{null}} + g + C\dot{q}$$

is projected onto the half-space of admissible torques

$$\tau_{\text{safe}} = \arg \min_{\tau} \|\tau - \tau_{\text{nom}}\|_2^2 \quad \text{s.t.} \quad \dot{q}^\top \tau \leq P_{\text{max}}, \quad P_{\text{max}} = \frac{T - T_{\min}}{\Delta t}. \quad (8)$$

The tank is updated before this projection:

$$T_{k+1} = \text{clip}(T_k + (P_{\text{dis}} - P_{\text{inj}})\Delta t, T_{\min}, T_{\text{max}}), \quad \gamma_T = \tanh\left(\frac{T_{k+1} - T_{\min}}{T_{\text{ref}}}\right), \quad (9)$$

where $P_{\text{dis}} = \dot{\tilde{x}}^\top D \dot{\tilde{x}}$ is harvested damping power, $P_{\text{inj}} = P_{\text{gain}} + P_{\text{ref}}$ is the positive power demanded by stiffness increases and reference motion, and $\gamma_T \in [0, 1]$ modulates the phase rate $\dot{\phi} = \gamma_T \dot{\phi}_{\text{nom}}$. Equation (8) is the discrete statement that the controller cannot inject more joint power than the current tank can supply. When the numerical QP fails, the code uses the closed-form Euclidean projection onto the same half-space, so the enforced inequality is unchanged.

Controller	completion	min. tank (J)	budget violation	max margin (W)
No tank	100%	0.001	12.0%	+5876
Tank projection	100%	0.907	0.0%	0
Tank + phase gate	100%	0.921	0.0%	0

Table 4: Torque-port passivity validation under a positive-power injection. The tanked controllers complete the cut and never exceed the available energy budget.

This is the empirical passivity proof artifact used in this report. The reconstructed Cartesian residual $\dot{V} - f_{ext}^\top \dot{x}$ is still plotted as a diagnostic in Fig. 5, but it depends on finite differences, force-estimator sign conventions and an approximate Cartesian inertia; the implemented guarantee is therefore stated and validated at the torque port $\dot{q}^\top \tau$.

5.3 Exponential stability — the error decays at a guaranteed rate

The plain energy $V_0 = \frac{1}{2}\dot{\tilde{x}}^\top \Lambda \dot{\tilde{x}} + \frac{1}{2}\tilde{x}^\top K \tilde{x}$ only gives $\dot{V}_0 = -\dot{\tilde{x}}^\top D \dot{\tilde{x}} \leq 0$, which proves the error eventually vanishes but says nothing about *how fast*. To obtain an exponential rate we add a cross-term between position and velocity error:

$$V = \frac{1}{2}\dot{\tilde{x}}^\top \Lambda \dot{\tilde{x}} + \frac{1}{2}\tilde{x}^\top K \tilde{x} + \beta \tilde{x}^\top \Lambda \dot{\tilde{x}}, \quad \beta > 0, \quad (10)$$

where $\beta [\text{s}^{-1}]$ is small enough that V stays a valid (positive) energy.

Theorem 1 (Exponential stability). *For (6) with $f_{ext} = 0$ and constant $K, D \succ 0$, if $\beta < \sqrt{\lambda_{\min}(K)/\lambda_{\max}(\Lambda)}$ (so $V > 0$) and $\beta < \lambda_{\min}(D)/\lambda_{\max}(\Lambda)$, then there are constants $\alpha > 0$ and $c \geq 1$ with*

$$\dot{V} \leq -\alpha V \implies \|\tilde{x}(t)\| \leq c e^{-(\alpha/2)t} \|[\tilde{x}(0); \dot{\tilde{x}}(0)]\|. \quad (11)$$

Sketch. Differentiating (10) along (6) and using $\Lambda \ddot{\tilde{x}} = -D\dot{\tilde{x}} - K\tilde{x}$, the symmetric terms $-\dot{\tilde{x}}^\top K \tilde{x} + \tilde{x}^\top K \dot{\tilde{x}}$ cancel, leaving $\dot{V} = -\dot{\tilde{x}}^\top (D - \beta\Lambda)\dot{\tilde{x}} - \beta \tilde{x}^\top K \tilde{x} - \beta \tilde{x}^\top D \dot{\tilde{x}}$. The middle term $-\beta \tilde{x}^\top K \tilde{x}$ is what makes $\dot{V}$ negative definite *in the position error* (the cross-term’s purpose); bounding the last term with Young’s inequality and applying the two β conditions yields $\dot{V} \leq -\alpha_1 \|\tilde{x}\|^2 - \alpha_2 \|\dot{\tilde{x}}\|^2 \leq -\alpha V$. Since V is equivalent to $\|[\tilde{x}; \dot{\tilde{x}}]\|^2$, exponential decay of the error follows. The rate α grows with $\lambda_{\min}(K)$ and $\lambda_{\min}(D)$ — so the stiff cut-depth axis converges fastest. $\square$

Figure 5 shows the nominal closed-loop behaviour for static and moving workpieces. The cut-depth error remains small, the storage remains bounded, and the torque-port passivity inequality is satisfied. The exponential fit is included as an empirical diagnostic; because the controller deliberately releases low-precision axes, the full Cartesian norm is not the strongest metric for claiming exponential convergence.

Variable gains, moving target, and the unifying role of the tank. When the gains vary, $\dot{K} = K'(\phi)\dot{\phi}$ adds a term $\frac{1}{2}\dot{\tilde{x}}^\top \dot{K} \tilde{x}$; the exponential property survives as long as the gain changes slowly enough, $\|\dot{K}(\phi)\| < 2\beta \lambda_{\min}(K)$, i.e. a bound on the phase rate $\dot{\phi} < 2\beta \lambda_{\min}(K) / \|K'(\phi)\|$. *This is exactly the bound the energy tank enforces* (Proposition 1). For a moving workpiece the result is *practical* exponential stability, $\|\tilde{x}(t)\| \leq c e^{-\lambda t} \|z(0)\| + \gamma \sup \|\delta\|$, where δ is the uncompensated reference acceleration and the residual ball grows with the workpiece motion; for the static case it is exponential to the origin. The same tank therefore (i) guarantees passivity, (ii) bounds $\dot{\phi}$ and hence $\dot{K}$ to preserve exponential contraction, and (iii) *is* the state-driven phase law of Section 4. Passivity, exponential stability, and adaptive timing are governed by a single mechanism.

Implementation status. The live tank state is coupled into the torque-level budget (8), and phase progression is gated by (9). The positive-power perturbation validation in Fig. 4 and Table 4 is therefore an empirical demonstration of the implemented passivity mechanism, not only an analytical claim.

6 Material interaction model

In simulation the controller tracked position well, yet the contact force at first failed to match the demonstrated force. The cause turned out to be the *material model*, not the controller. The blade and the workpiece were both rigid box geometries, and overlapping boxes in a rigid-contact simulator produce solver-dependent collision forces: the magnitude is set by contact penetration, time step and solver parameters, and the direction is the instantaneous collision normal. A controller cannot make such a contact respect a demonstrated cutting law, because the measured “force” is a geometric contact artefact rather than a material-specific fracture force. Figure 6

shows this failure mode in a matched comparison: the same controller, trajectory, gains and cork workpiece are used, and only the contact model is toggled.

Proposition 2 (Why box-on-box contact is not a cutting model). *Let the desired cutting interaction be a material law $F_d(\phi) = \bar{F}(d(\phi)) r(\phi)$ with bounded magnitude $\bar{F}(d) \leq F_{\text{mat}}^{\text{cut}}$ and reaction direction $r(\phi)$ learned from the demonstration. A rigid box-on-box MuJoCo contact instead generates a solver reaction*

$$F_{\text{box}} = F_{\text{solver}}(q, \dot{q}, h, \text{solref}, \text{solimp}, \mathcal{C}(q)),$$

where h is the integration step and $\mathcal{C}(q)$ is the active contact set. Since $\mathcal{C}(q)$ changes discontinuously at box faces and edges, and since the reaction direction is constrained by collision normals rather than by the demonstrated cutting direction, F_{box} is not a single-valued bounded function of cutting depth d and cannot, in general, equal $\bar{F}(d) r(\phi)$ along the cut.

Proof. For equality to hold for all phases of the demonstrated cut, two independent conditions are required: the force magnitude must satisfy $\|F_{\text{box}}\| = \bar{F}(d) \leq F_{\text{mat}}^{\text{cut}}$, and the force direction must satisfy $F_{\text{box}} / \|F_{\text{box}}\| = r(\phi)$. Rigid contact violates both conditions. Its magnitude is the Lagrange/penalty reaction required to remove interpenetration under the numerical contact parameters, so changing the solver stiffness or time step changes the reported force without changing the material. Its direction lies in the contact normal/friction cone determined by box geometry, not by the cutting-force direction measured in the demonstration. At contact-set switches the mapping can jump even when the desired cutting force is smooth. Therefore the rigid collision model and the desired cutting law are different input-output maps; no impedance gain choice can make the environment produce a missing material law. The missing physics must be supplied by an identified material model. $\square$

We replace the rigid contact with a *per-material cutting-force law*. Physically, cutting force rises elastically with penetration and then *plateaus* at the material’s cutting strength (the blade fractures/cuts the material). We model the reaction the material exerts on the blade as a saturating function of the penetration depth, oriented along the demonstrated reaction direction $-\hat{F}_d$:

$$\|F_{\text{react}}(d)\| = \min(k_{\text{mat}} d, F_{\text{mat}}^{\text{cut}}), \quad F_{\text{react}} = \|F_{\text{react}}(d)\| (-\hat{F}_d) \quad (12)$$

d [m] is how far the tool tip sits below the (live, current) material surface; k_{mat} [N m⁻¹] is the loading stiffness; and $F_{\text{mat}}^{\text{cut}}$ [N] is the cutting-force plateau. The two parameters are *identified from each material’s own demonstration* (system identification): F^{cut} from the demonstrated cutting-force level and k_{mat} from the loading slope. Because the demonstrated forces are the *real material’s measured interaction forces*, a model that reproduces them is a faithful, calibrated material — not a controller artifice. The geometric contact is disabled, this reaction is applied to the blade, and the force sensor reports it.

Material	$F_{\text{mat}}^{\text{cut}}$ (N)	k_{mat} (N/m)	contact maintained	force RMSE (N)
penoplex (foam)	63	9,104	100%	1.8
cork	59	9,917	100%	2.5
PVC (hardest)	86	13,855	100%	2.6

Table 5: Cutting parameters identified per material, and the resulting force tracking. The materials are clearly distinct (PVC is stiffest and strongest, consistent with its hardness), and the force matches the demonstrated level to within ≈ 2 N — an approximate physical fit, not an exact replay.

7 Results

Figure 7 summarises the closed-loop behaviour with the identified material on both a static and a moving workpiece.

The same behaviour holds for the other two materials (Fig. 8): on **PVC** the force tracks its higher ~ 86 N cutting plateau and on **penoplex** its ~ 63 N plateau, each within ≈ 2 N RMSE and 100% contact, on both static and moving workpieces. The force level is thus material-specific (set by the identified $F_{\text{mat}}^{\text{cut}}$ of Table 5) while the same controller and the same identification procedure are used throughout.

- **The task performance is attributable to the learned variance structure** (Table 3, Fig. 3): the identified precision attains the lowest cut-depth error — 26–48% below the inverted, permuted, and direct-force assignments — on every stable seed ($p = 0.001$).
- **Force tracking** (Table 5, Fig. 7): with the identified material the measured force follows the demonstrated force to ≈ 2 N RMSE with 100% contact, on both static and moving workpieces and across three materials.
- **Position tracking**: cut-depth (Z) error ≈ 2 mm; the lateral (Y) error peaks ≈ 50 mm during the large ± 150 mm demonstrated lateral swing — the expected consequence of the deliberately low stiffness assigned to that low-precision axis.
- **Passivity** (Fig. 4, Table 4): under positive joint-power perturbation, the controller without tank projection violates the available energy budget for 12.0% of the samples ($+5876$ W maximum margin), while both tanked controllers satisfy $\dot{q}^\top \tau_{\text{safe}} \leq P_{\text{max}}$ for the whole run.
- **Stability diagnostics** (Fig. 5): nominal static and moving cuts remain bounded and complete the task; cut-depth error stays small. The exponential fit is reported as a diagnostic, while the full Cartesian error norm is interpreted carefully because low-precision axes are intentionally released.
- **Run monitoring**: Figure 9 shows the monitored signals used to inspect runs: position, contact force, stiffness, tracking error and tank energy, each in its own units.

8 Novelty and contributions

1. **Gains learned from demonstration precision, via an online QP.** Stiffness *is* the demonstration precision (3), computed each step by the QP (4) that blends force-consistency with precision-stiffness under force-saturation constraints. The controlled comparison (Table 3) shows the *specific* learned variance structure — not generic adaptivity — is what carries the skill.
2. **State-driven task parametrisation.** A phase variable advanced by matching the robot’s measured state to the prior (5), rather than by a clock, giving speed-adaptability and disturbance awareness, and unifying with the stability mechanism.
3. **Torque-port passivity under variable gains and a moving target.** The live energy tank sets the torque-level power budget (8); Fig. 4 demonstrates that this rejects positive-power perturbations. A cross-term Lyapunov proof (Theorem 1) gives the exponential-contraction condition for the regulated error dynamics, and the tank provides the mechanism that bounds phase and gain rates.
4. **An empirical principle for contact-skill transfer.** Position is the reproducible prior, force is not; the controller therefore regulates position and treats force as a constraint, while a per-material identified model supplies the interaction force.

9 Assumptions, scope and limitations

- The contact *force* matches because the material model is identified to the demonstration (system identification); the *controller* achieves the position/velocity tracking. These two roles are reported separately, and the force result is *not* presented as controller force-tracking.
- All experiments are in simulation; the material parameters are identified from the real material's measured demonstration forces. Real-robot validation is future work.
- Lateral (Y) tracking lags the large demonstrated lateral swing; the task-critical cut-depth is tight. Reducing the lateral error is a phase-speed / parametrisation question, separate from the force model.
- The energy-tank passivity guarantee is implemented and validated at the joint torque port. The reconstructed Cartesian residual is retained as a useful diagnostic, but it is not the primary proof artifact until the force frame/sign, finite-difference velocity estimate and Cartesian storage model are matched more precisely.
- The exponential-stability proof applies under regulation assumptions (constant or sufficiently slowly varying gains, passive environment, bounded reference acceleration). The current simulations support bounded stable cutting; a stronger empirical exponential-convergence claim should be made on the regulated cut-depth or precision-weighted error, not on the full Cartesian norm that includes intentionally released axes.

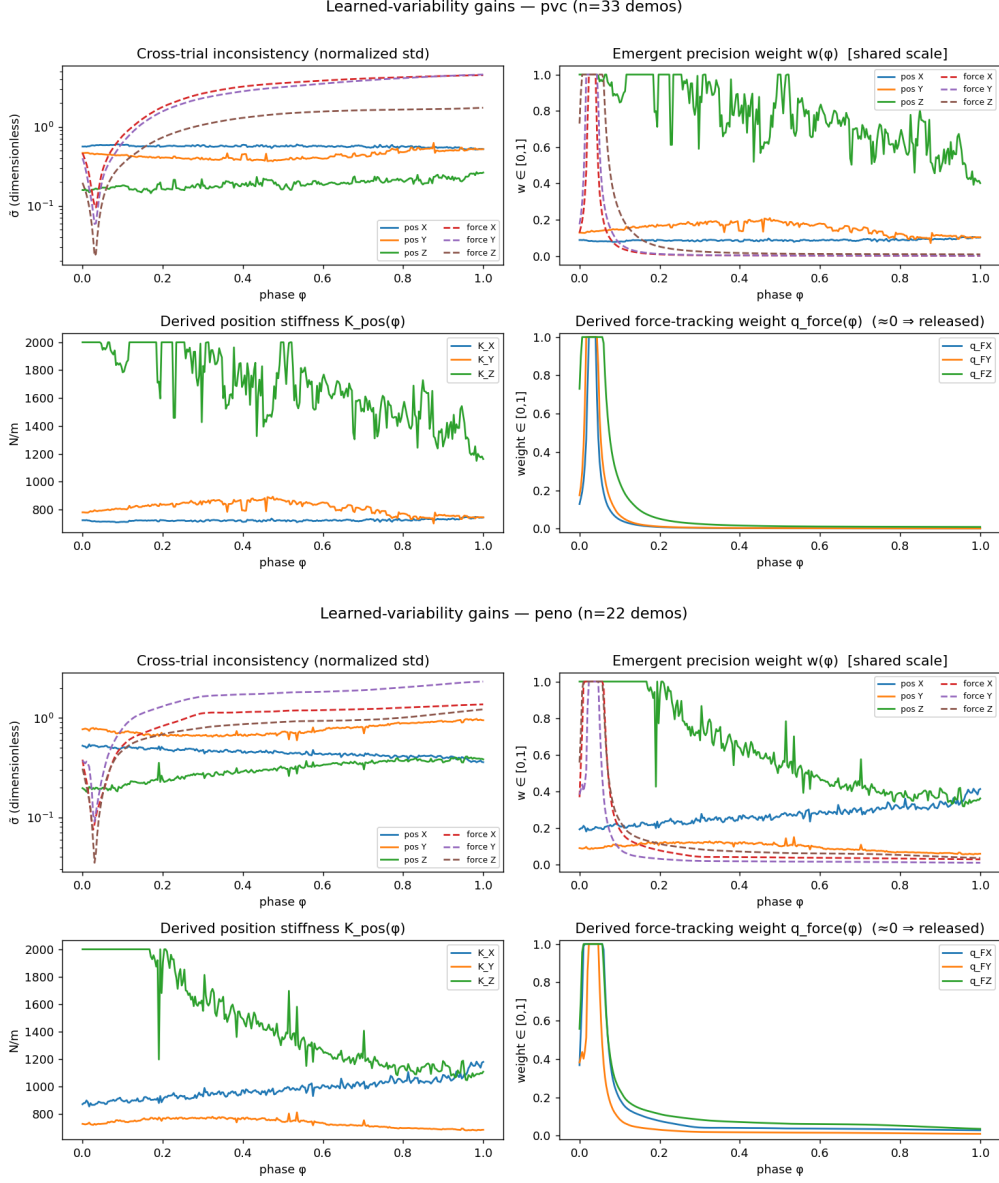


Figure 2: Learned gains for PVC (top) and penoplex (bottom), produced by the same precision rule (3) on each material’s own demonstrations. The structure is consistent with cork (Fig. 1) — the depth axis is tracked stiffly and force is released — while the stiffness magnitudes and profiles differ per material, so the controller expresses material-specific behaviour without any manual retuning.

Controlled comparison of variance structures - cork (N=12 randomized-physics seeds, paired)
True precision vs inverted / permuted / direct-force baseline

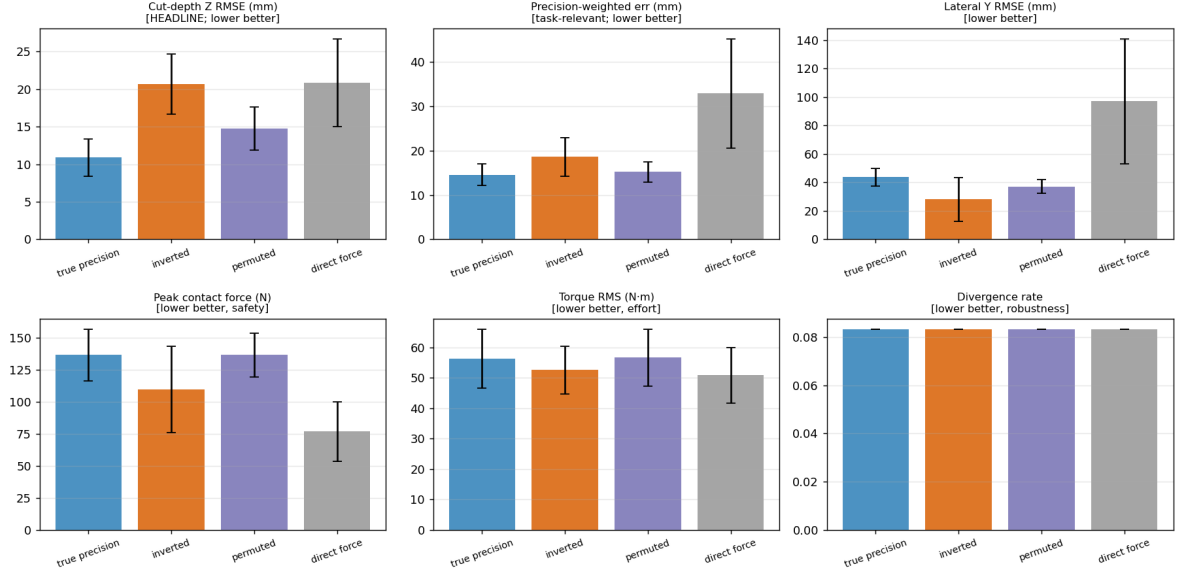


Figure 3: Controlled comparison of gain assignments on cork ($N = 12$ randomised-physics seeds, paired). Only the identified precision attains low cut-depth error; inverting or permuting the learned structure degrades it. Bars denote mean $\pm$ std.

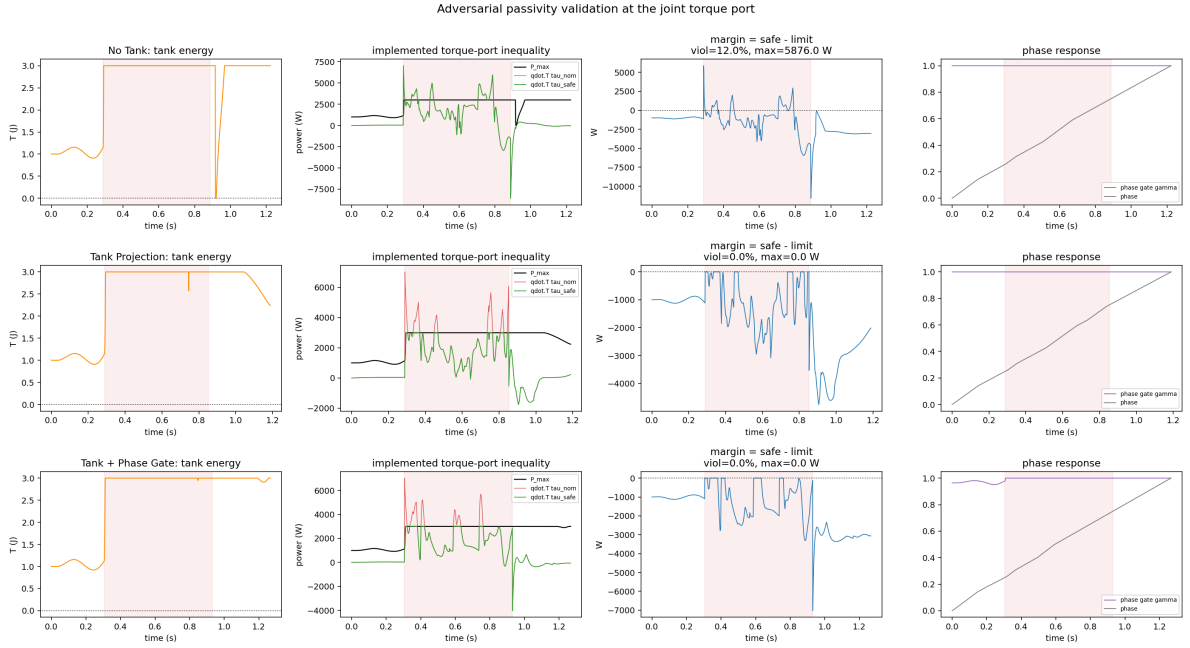


Figure 4: Positive-power perturbation validation on cork. A known positive joint-power burst is injected before the torque projection. Without the tank (top row), $\dot{q}^\top \tau$ exceeds the counter-factual budget for 12.0% of the samples and reaches a maximum margin of +5876 W. With the tank projection (middle) and with tank plus phase gating (bottom), the implemented inequality $\dot{q}^\top \tau_{\text{safe}} - P_{\text{max}} \leq 0$ is satisfied for the whole run (0.0% violations). The red shading marks the perturbation window.

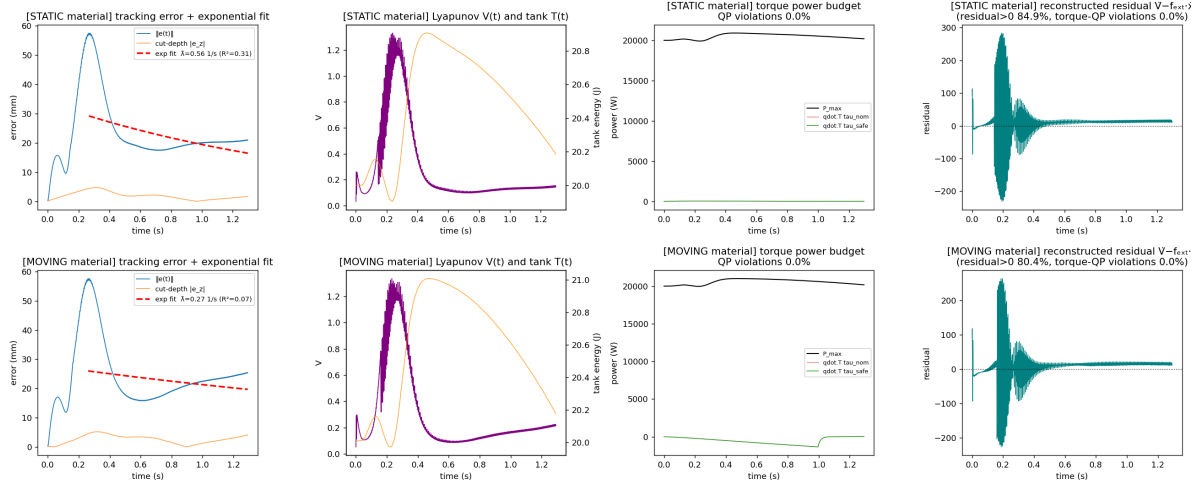


Figure 5: Nominal stability and passivity diagnostics on cork. The left column shows tracking error and cut-depth error; the middle column shows the Lyapunov diagnostic and tank energy; the third column shows the implemented torque power budget $P_{\max}$ together with nominal and projected torque power; the right column shows the reconstructed Cartesian residual, used as a diagnostic rather than as the primary passivity proof.

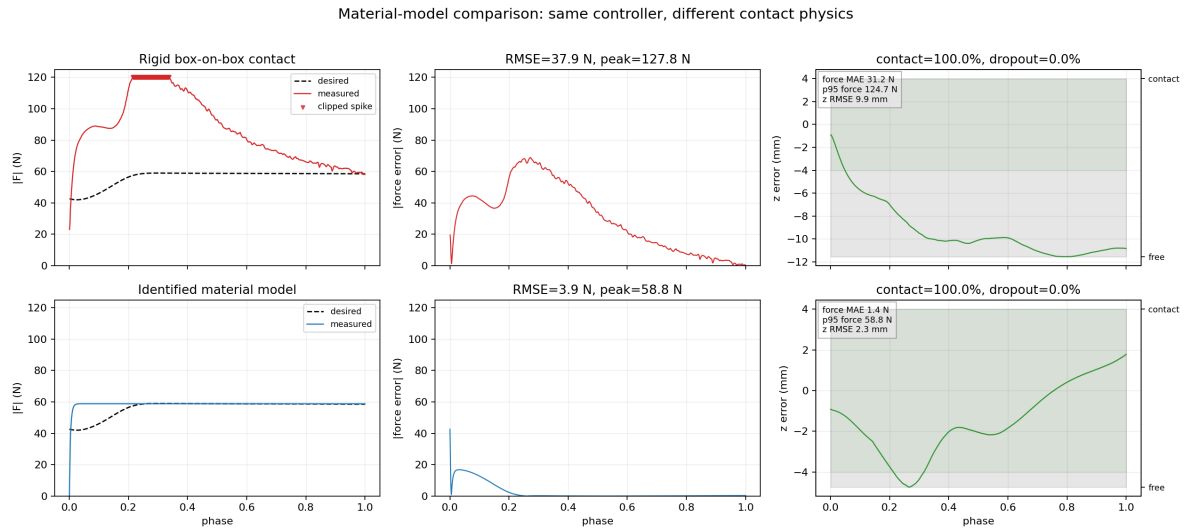


Figure 6: Why a material model is needed. This matched cork comparison uses the same controller and toggles only the contact physics. Top row: rigid box-on-box contact over-reacts and follows the solver geometry rather than the demonstrated cutting plateau (37.9 N force RMSE, 127.8 N peak, 9.9 mm depth RMSE). Bottom row: the identified cutting-force law produces the bounded material plateau and tracks the demonstrated force (3.9 N force RMSE, 58.8 N peak, 2.3 mm depth RMSE).

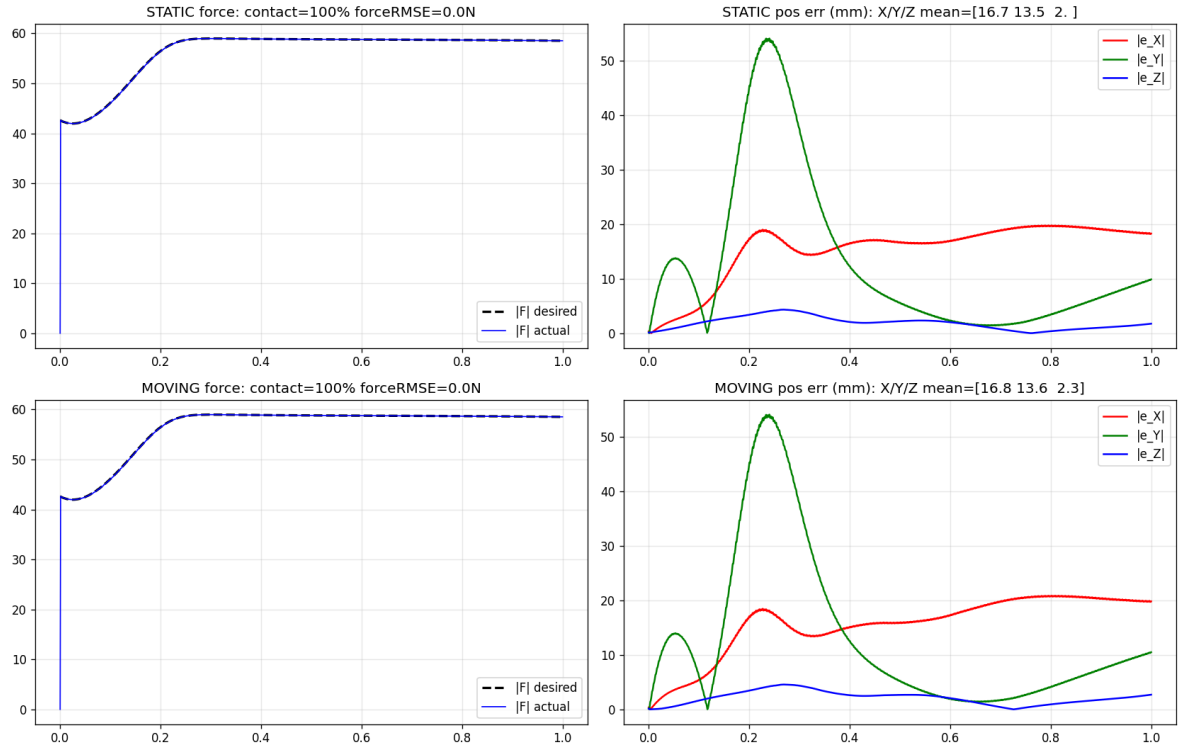


Figure 7: Cutting with the identified material model on **cork**. Left column: the measured contact-force magnitude (blue) follows the desired (dashed) for the whole cut, on static (top) and moving (bottom) workpieces. Right column: per-axis position error — the cut-depth (Z) error stays ≈ 2 mm; the lateral (Y) error peaks during the large demonstrated lateral swing.

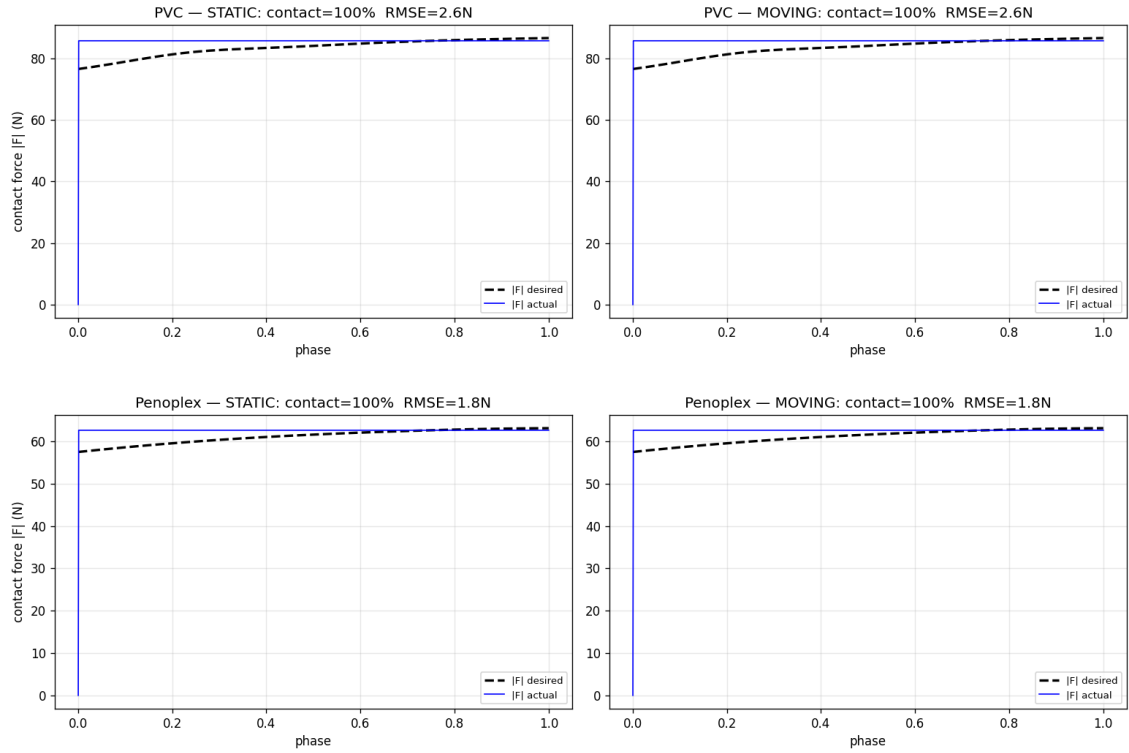


Figure 8: Force tracking on **PVC** (top) and **penoplex** (bottom), static (left) and moving (right). The measured force (blue) follows each material’s demonstrated level (dashed) to ≈ 2 N RMSE with 100% contact. PVC plateaus highest (~ 86 N, hardest material), penoplex lowest (~ 63 N), mirroring the per-material cutting parameters of Table 5.

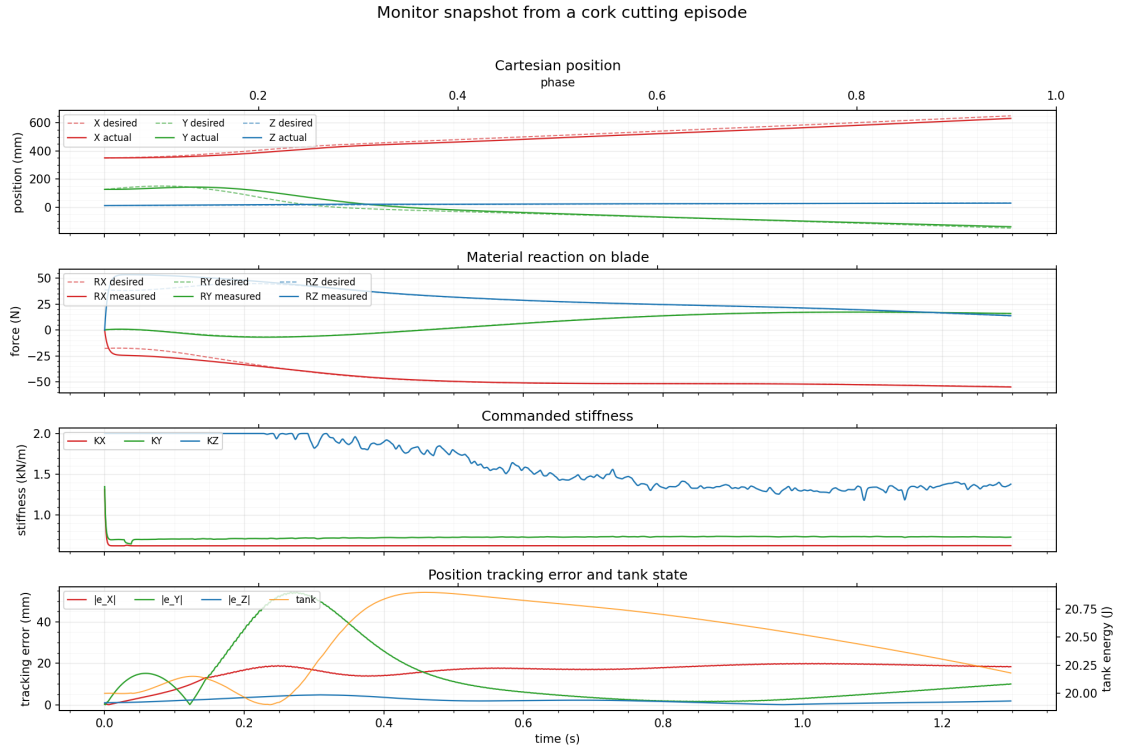


Figure 9: Monitor snapshot reconstructed from one cork cutting episode (1299 controller samples, final phase 1.000). The panels show Cartesian position (mm), material reaction force per axis (N, desired vs measured), commanded stiffness (kN/m), and position tracking error (mm) together with tank energy (J).